

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 128

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

This demonstrates that passive interfaces prevent OSPF neighbor adjacencies from forming on those specific interfaces while still allowing the network to be advertised. R2 can only form OSPF adjacencies on interfaces that are not passive.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

Cost of S0/0/1 on R3 + Cost of G0/0 on R2 = 64 + 1 + 65

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

By increasing the reference bandwidth (e.g., to 10000 for 10 Gbps), OSPF can accurately calculate different costs for different speed links, allowing for better path selection.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For 192.168.3.0/24, Cost = $100000/128$ (S0/0/1 on R1) + 1 (G0/0 on R3) = 781 + 1 = 782

For 192.168.23.0/30, Cost = $100000/128$ (S0/0/0 on R1) + $100000/1544$ (G0/0 on R2/R3) = 781 + 64 = 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

After applying the bandwidth 128 command on all remaining serial interfaces, both serial links from R1 to R2 and R1 to R3 have a cost of 781 each.

Total: $781 + 781 = 1562$

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Path through R2: Cost = $781 + 781 + 1 = 1563$

Path through R3: Cost = $1565 + 1 = 1566$

Since OSPF always choose the path with the lowes cost, R1 now routes to the 192.168.0/24 network through R2 with cost 1563 instead of the direct path through R3 with cost 1566.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Because a manually chosen router ID guarantees each router has a unique, consistent identifier that does not change when interfaces or IP addresses change, which simplifies troubleshooting, log reading, and DR/BDR elections across the OSPF domain.

2. Why is the DR/BDR election process not a concern in this lab?

Because OSPF only performs DR/BDR elections on multi-access networks with several routers on the same segment, and in this lab all OSPF adjacencies are on point-to-point serial links, while the Ethernet interfaces connect only to PCs, not to other routers.

3. Why would you want to set an OSPF interface to passive?

Set an interface to passive when no OSPF neighbor should ever form on that link (for example, a LAN to end users), so that OSPF still advertises the connected network in the LSDB but stops sending hello packets and routing updates, reducing protocol overhead and limiting exposure to rogue OSPF neighbors.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF's use of multiple areas with a required backbone area 0, plus Area Border Routers that summarize and restrict LSAs between areas, creates a hierarchy that keeps each area's link-state database small, reduces SPF calculations, and allows very large topologies to converge efficiently.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

Command: network 10.0.0.0 0.255.255.255 area 0

This wildcard mask (0.255.255.255) matches every interface whose IP is in the 10.0.0.0/8 block, so all such interfaces join area 0 with one statement.

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route is used, because the router first compares administrative distance and always prefers the protocol with the lowest administrative distance, regardless of each protocol's internal metric value.